CHEVRON CORPORATION

INTERNAL USE ONLY

MOC Digital Application

Management of Change — HSE Digital Platform

DOCUMENT DATE	VERSION	STATUS
April 26, 2026	2.0	Demo / Prototype

Table of Contents

1 Document Overview & Purpose
2 Demo Users & Roles
3 8-Step Workflow Summary
4 Step-by-Step Demo Walkthrough
Step 1 — Login
Step 2 — Admin: The Big Picture
Step 3 — Owner: Create a New MOC
Step 4 — MOC Detail View
Step 5 — Evaluator: Complete the Evaluation
Step 6 — Owner: Submit for Approval to Implement
Step 7 — Manager: Approve or Reject
Step 7a — Rejection & Owner Revision Flow
Step 8 — Owner: Implementation
Step 9 — PSSR (Pre-Startup Safety Review)
Step 10 — Manager: Approve Start Up
Step 11 — Owner: Ready for Closure
Step 12 — Notifications
Step 13 — Closed MOC: Full Audit Trail
Step 14 — Assigned Tasks & Your Status
5 Action Item Types & Further Action Items
6 Role & Visibility Matrix
7 Key Rules & Guardrails
8 Technology Stack

1. Document Overview & Purpose

The **Chevron MOC Digital Application** is a Management of Change (MOC) platform built for Chevron's HSE Digital initiative. It provides a structured, auditable workflow for proposing, evaluating, approving, implementing, and closing engineering and operational changes in compliance with Chevron's HSE standards.

The application replaces paper-based MOC processes with a fully digital, role-based system that enforces stage-gate controls, tracks all decisions with timestamps, and maintains an immutable audit trail from initiation through closure.

This document serves as the official demo guide and functional reference for the application prototype.

Quick Start

STEP	COMMAND	NOTES
1. Install dependencies	`npm install`	One time only
2. Build the app	`npx ng build`	Compiles Angular to `dist/`
3. Start the server	`node serve-dist.js`	Serves on port 4300
4. Open in browser	`http://127.0.0.1:4300/`	Demo password: **demo123**

2. Demo Users & Roles

Use **Persona Login** (one-click) or **ID & Password** with password `demo123` for any user.

LOGIN NAME	ROLE	DISCIPLINE	PRIMARY RESPONSIBILITIES
Olivia	Owner	—	Creates MOCs, submits for approval, drives implementation, closes MOC
Owen	Owner	—	Secondary Owner for demonstration variety
Mike	Evaluator	Mechanical	Completes Mechanical discipline evaluation checklist
Priya	Evaluator	Process	Completes Process discipline evaluation checklist
Evan	Evaluator	Environmental	Completes Environmental evaluation checklist
Harper	Evaluator	HSE	Completes HSE evaluation checklist
Oscar	Evaluator	Operations	Completes PSSR (Pre-Startup Safety Review) checklist
Morgan	Manager	—	Approves or rejects at Approval to Implement and Approval for Start Up gates
Ada	Admin	—	Views all MOCs, configures checklists, can reset demo data

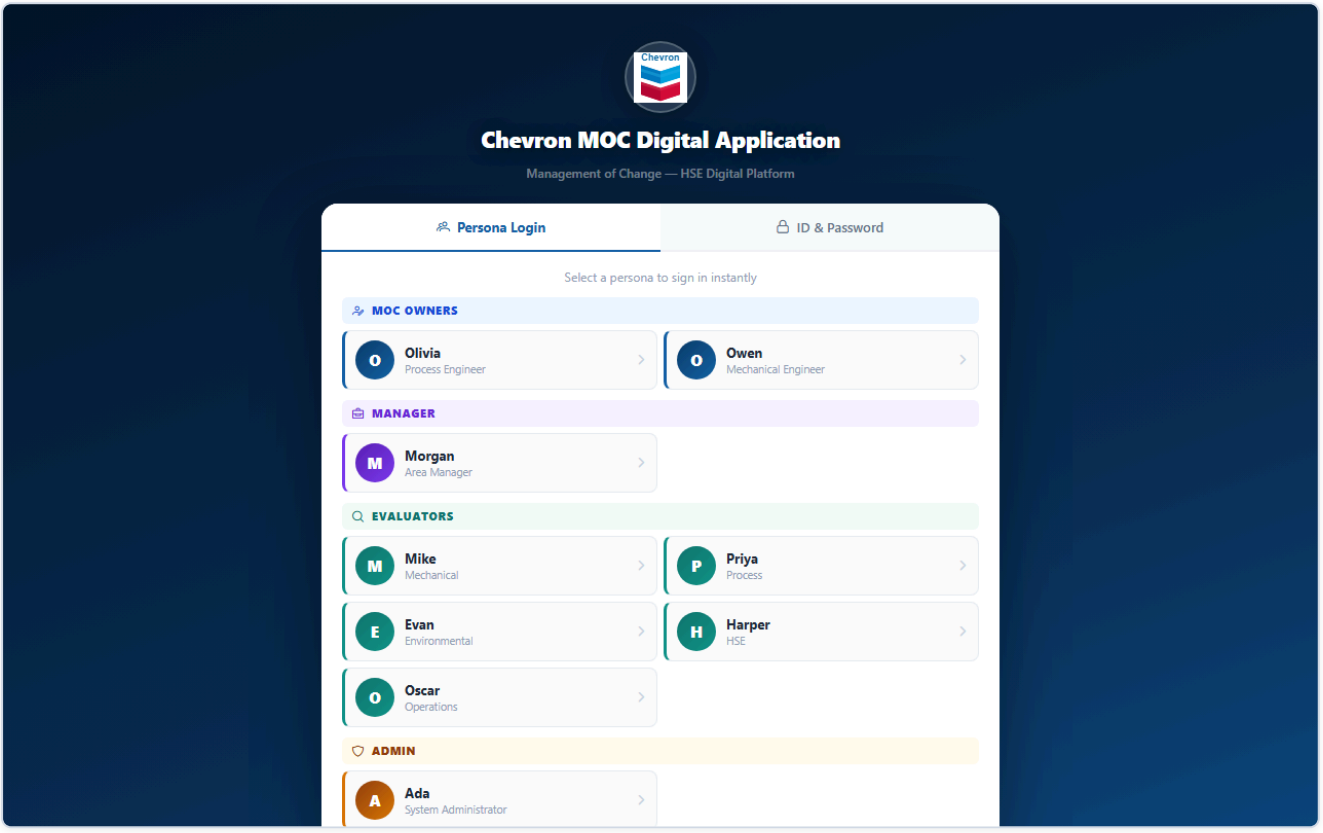


Fig 1 — Persona Login: click any user tile to sign in instantly

3. 8-Step Workflow Summary

#	STAGE	ROLE	WHAT HAPPENS	GATE CONDITION
1	Evaluation	Owner	Owner fills in change details, selects disciplines, uploads supporting document, and submits. MOC auto-routes to all assigned Evaluators.	All fields filled
2	Evaluation	Evaluators	Each discipline Evaluator independently completes their checklist and raises action items (Evaluation / Pre-Startup / Post-Startup phase). All Evaluation AIs must be closed before Owner can advance.	All evals + Eval AIs complete
3	Approval to Implement	Owner → Manager	Owner submits for approval. Manager reviews and approves (→ Implementation) or rejects (→ Rejected state, Owner revises).	Manager decision
	Rejected	Owner	Manager rejected: Owner sees rejection reason, edits initiation fields, and resubmits. Revision History records all field changes. All evaluations reset.	Owner resubmits
4	Implementation	Owner	Owner marks implementation complete after all field work, procedure updates, and drawing revisions are done.	Owner confirms complete
5	PSSR	Operations	Operations completes PSSR checklist and closes all Pre-Startup action items. Operations may also raise new Pre-Startup / Post-Startup AIs during this stage.	PSSR checklist + Pre-Startup AIs
6	Approval for Start Up	Manager	Manager reviews PSSR results and approves startup (→ Ready for Closure) or rejects back to PSSR for additional review.	Manager decision
7	Ready for Closure	Assignees → Owner	All Post-Startup action items must be closed (each with a comment and optional evidence). Only then can Owner close the MOC.	All Post-Startup AIs closed

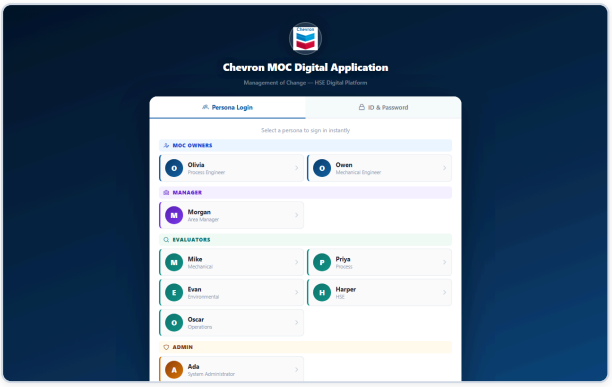
#	STAGE	ROLE	WHAT HAPPENS	GATE CONDITION
8	Closed	Owner	MOC permanently closed. Full audit trail (evaluations, checklists, action items, approvals, revision history) locked and read-only.	Final state

4. Step-by-Step Demo Walkthrough

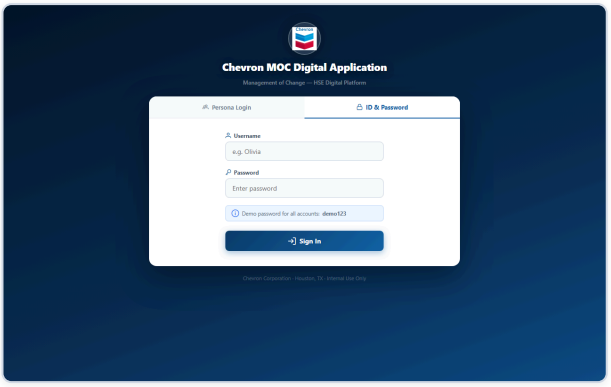
1 Login

All Users

Two login modes are available: **Persona Login** (click a user tile — instant sign-in) and **ID & Password** (username + `demo123`). Use Persona Login for live demos.



Persona Login — click any tile to sign in instantly

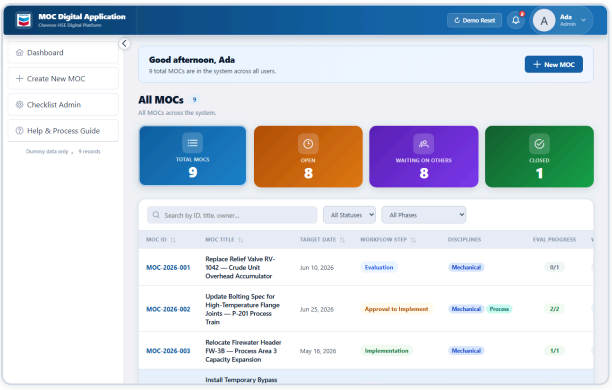


ID & Password tab — username + demo123

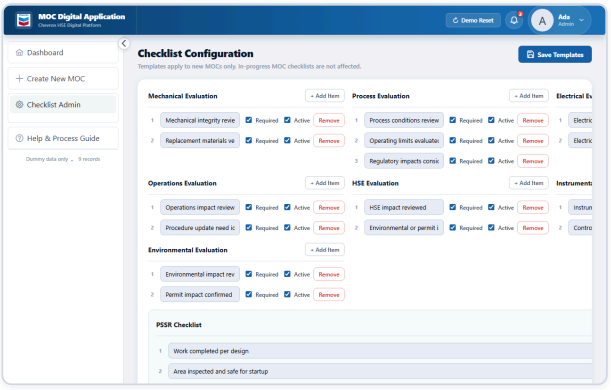
2 Admin: The Big Picture

Role: Ada (Admin)

Admin sees every MOC across all users. The Demo Reset button restores seed data at any time. Admin also configures discipline evaluation checklists and the PSSR checklist.



Admin dashboard — all MOCs in the system



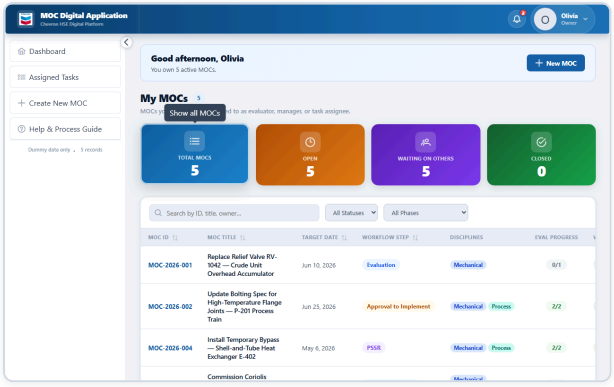
Checklist Admin — configure evaluation checklist items per discipline

3 Owner: Create a New MOC

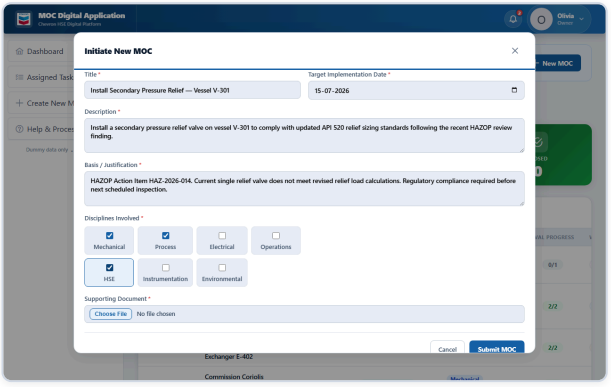
Role: Olivia (Owner)

Click **+ New MOC** to open the initiation form. Fill in all required fields and click **Submit MOC** — the record is created and immediately routed to assigned Evaluators.

- **Title** — short descriptive name
- **Description** — what the change is
- **Basis / Justification** — HAZOP finding, regulatory requirement, etc.
- **Disciplines Involved** — each selected discipline gets its own evaluator
- **Supporting Document** — engineering basis or HAZOP report
- **Target Implementation Date** — when the change must be in place



Owner dashboard — My MOCs with KPI summary tiles

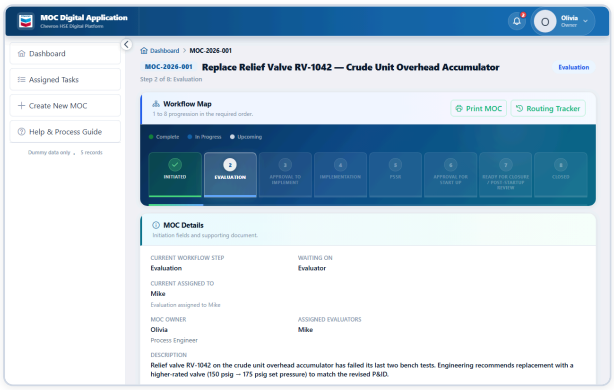


Create MOC modal — filled out and ready to submit

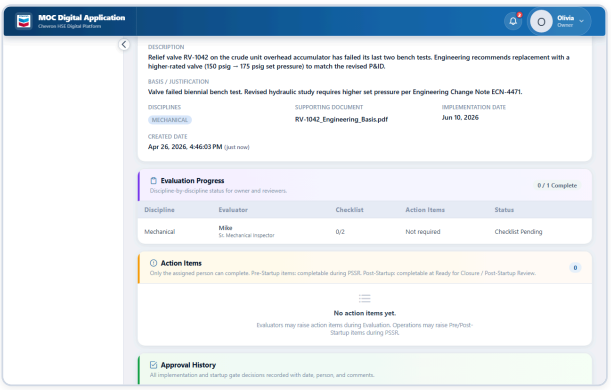
4 MOC Detail View

All Users

Click any MOC row to open the detail view — the single source of truth for the entire lifecycle. The 8-step workflow map at the top shows current position with completed steps checked.



Workflow map — visual 8-step progress indicator



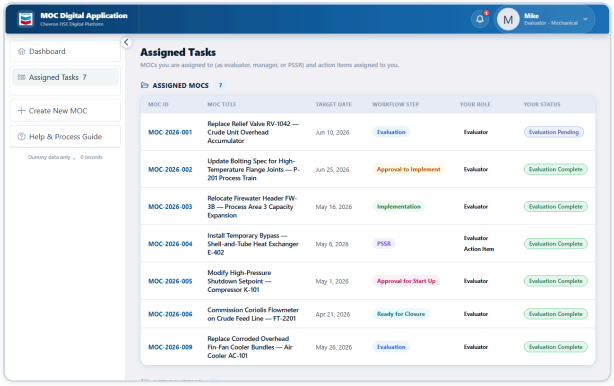
Evaluation Progress — per-discipline checklist and action item status

5 Evaluator: Complete the Evaluation

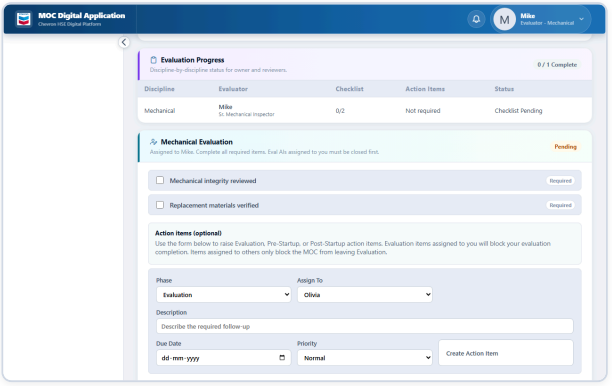
Role: Mike (Evaluator — Mechanical)

Sign in as Mike and go to **Assigned Tasks**. Click any MOC to open the evaluation panel for the Mechanical discipline.

- Complete all required checklist items
- Raise action items (choose Evaluation / Pre-Startup / Post-Startup phase)
- Close your own assigned action items with a closure comment
- Click **Mark Evaluation Complete**



Evaluator Assigned Tasks — all assigned MOCs and action items

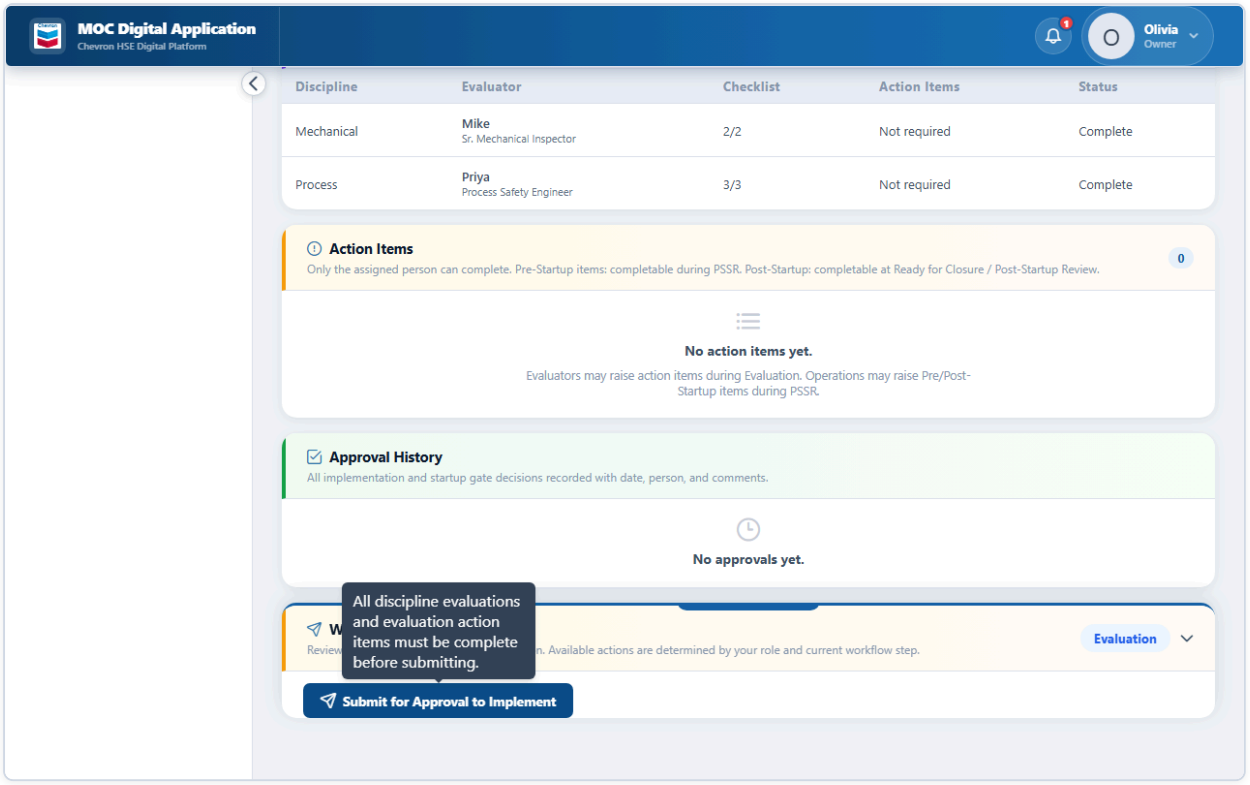


Evaluation checklist, action item form, and Mark Evaluation Complete

6 Owner: Submit for Approval to Implement

Role: Olivia (Owner)

Once all discipline evaluations and Evaluation-phase action items are complete, the Owner sees the **Submit for Approval to Implement** button in the sticky Workflow Actions panel at the bottom of the MOC detail.



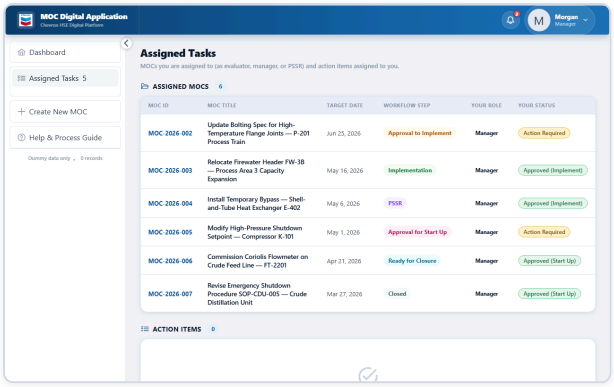
Workflow Actions panel — Submit for Approval to Implement button

7 Manager: Approve or Reject

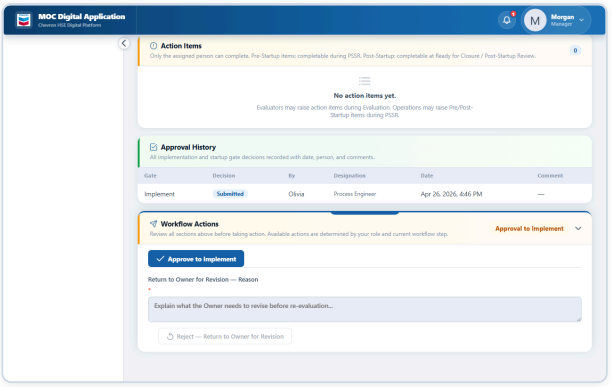
Role: Morgan (Manager)

Sign in as Morgan. **Assigned Tasks** shows all MOCs awaiting Manager action (Action Required) and the full history of past approvals. Click any MOC, review the full detail, scroll to Workflow Actions.

- **If Approved** — MOC moves to Implementation; Owner is notified
- **If Rejected** — Manager must enter a reason; MOC enters *Rejected* state and routes to Owner for revision



Manager Assigned Tasks — full history with Your Status column



Manager Workflow Actions — Approve or Reject (Return to Owner)

7a Rejection & Owner Revision Flow

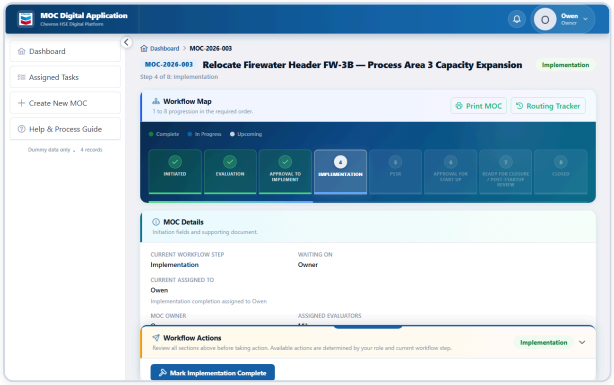
Role: Owner (after rejection)

↩ When a Manager rejects, the MOC enters **Rejected** state (shown at the Initiated position on the workflow map). The Owner sees a red Rejection Reason box with the Manager's name, date, and comment. The Owner clicks **Revise MOC Details**, edits any initiation field, and clicks **Submit Revised MOC**. The MOC re-enters Evaluation with all discipline evaluations reset. A **Revision History** block records every field change as an old → new diff. This cycle can repeat as needed — all history is permanently preserved.

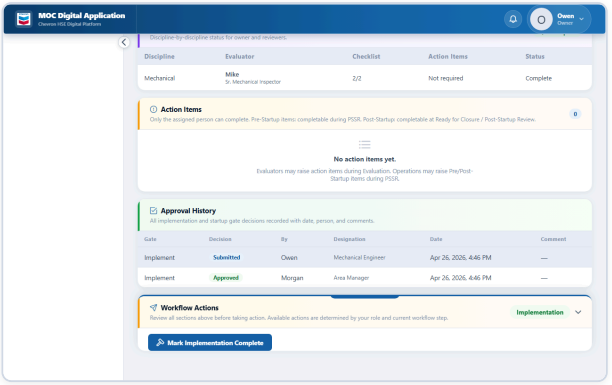
8 Owner: Implementation

Role: Owen (Owner)

After Manager approval the Owner drives field implementation. Click **Mark Implementation Complete** to confirm all physical changes, procedure updates, and drawing revisions are done.



MOC in Implementation — approval history shows Manager approved

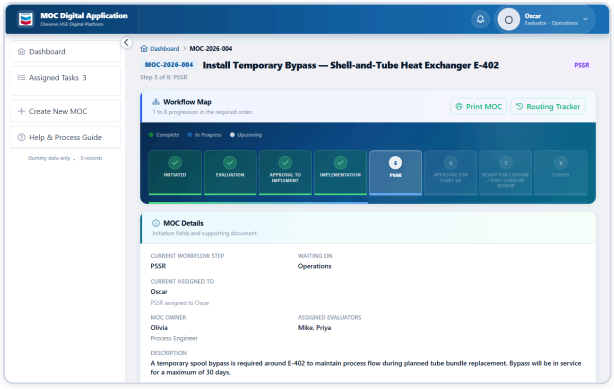


Workflow Actions — Mark Implementation Complete button

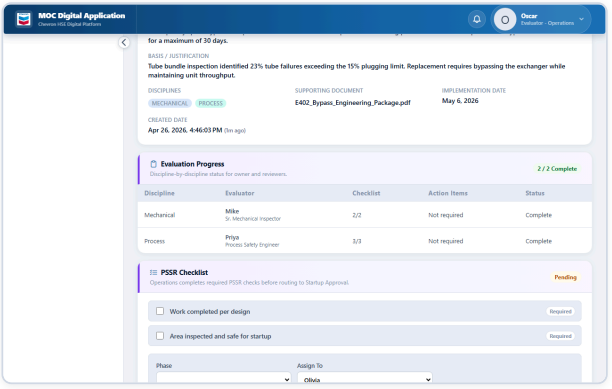
9 PSSR — Pre-Startup Safety Review

Role: Oscar (Evaluator — Operations)

Operations completes the PSSR checklist and closes all Pre-Startup action items. Operations can also raise new Pre-Startup or Post-Startup action items during this stage.



PSSR MOC Detail — workflow map at PSSR step

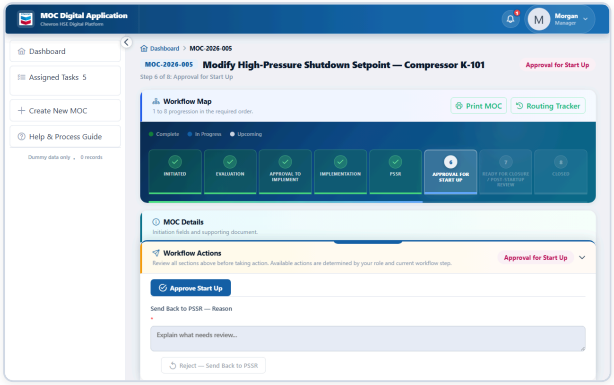


PSSR checklist — all items must be checked before submission

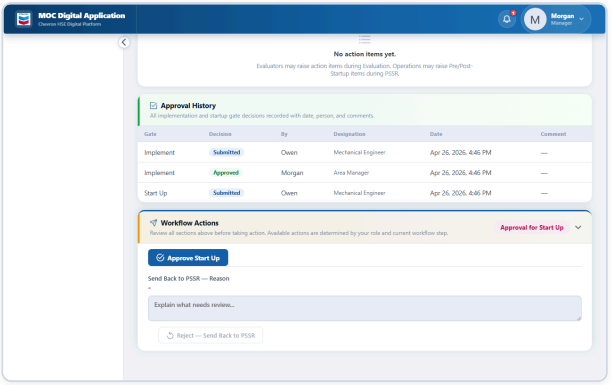
10 Manager: Approve Start Up

Role: Morgan (Manager)

Manager reviews PSSR outcome and approves or rejects startup. All decisions recorded in Approval History.



Manager reviews PSSR results before approving startup

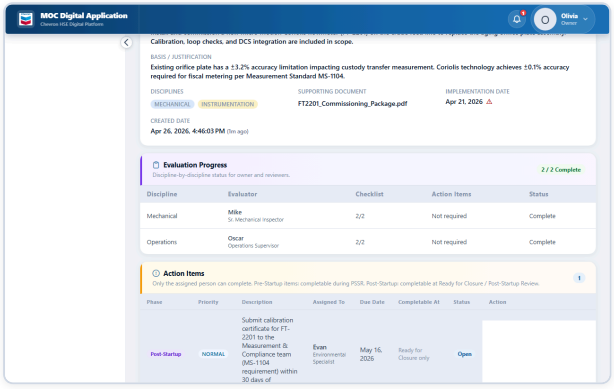


Workflow Actions — Approve Start Up or Reject (Send Back to PSSR)

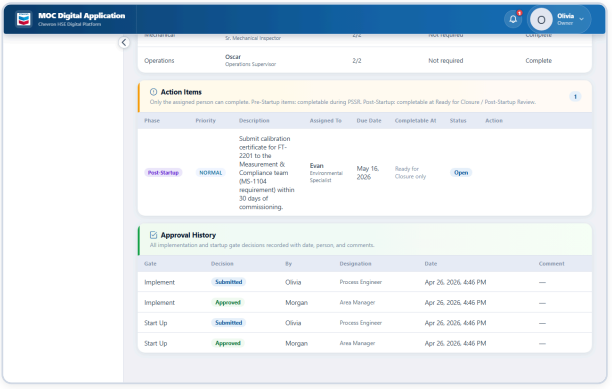
11 Owner: Ready for Closure

Role: Olivia (Owner)

All Post-Startup action items must be closed (each with a comment and optional evidence) before the **Close MOC** button activates.



Post-Startup action items — close each with a comment

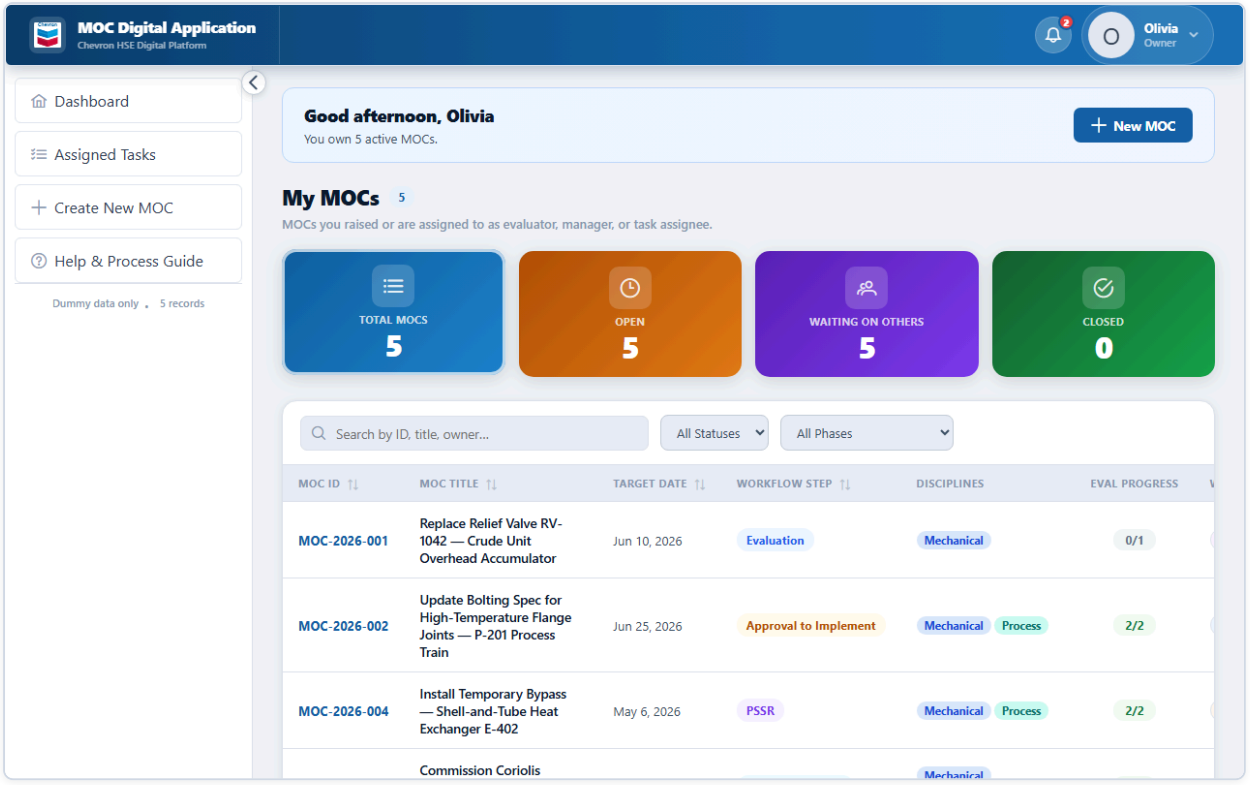


Close MOC button — enabled only when all post-startup items are done

12 Notifications

All Users

The bell icon in the topbar shows unread notifications. Each workflow transition generates a contextual notification for the relevant user. Read-state persists across sessions.

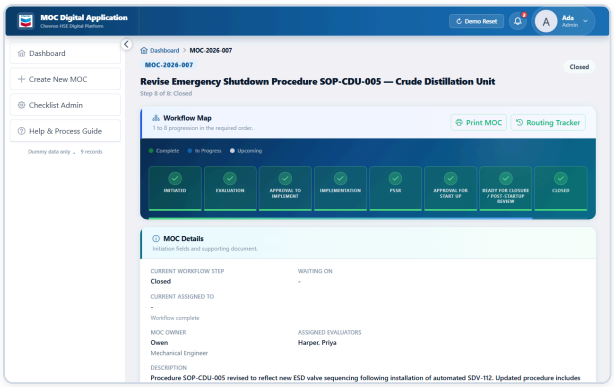


Notification panel — workflow-state-specific messages with mark-read controls

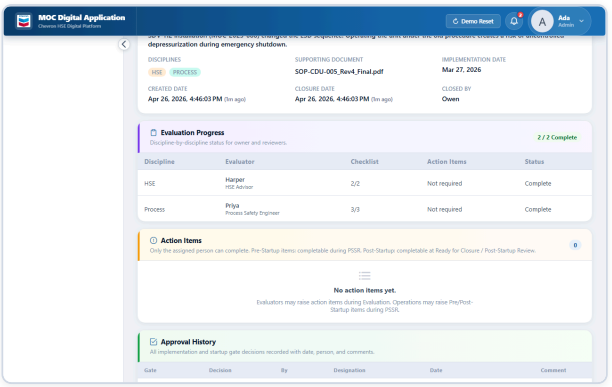
13 Closed MOC: Full Audit Trail

Role: Ada (Admin)

As Admin, open any closed MOC to see the complete permanent record — all sections are read-only. Every decision, comment, checklist response, and action item is preserved with timestamps.



Closed MOC detail — read-only, full history preserved



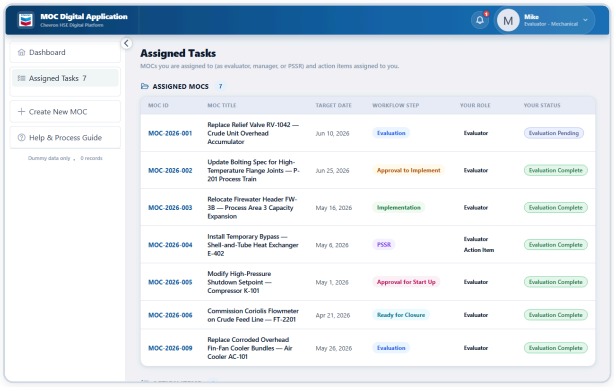
Approval History — every Manager decision with timestamp and comment

14 Assigned Tasks & Your Status

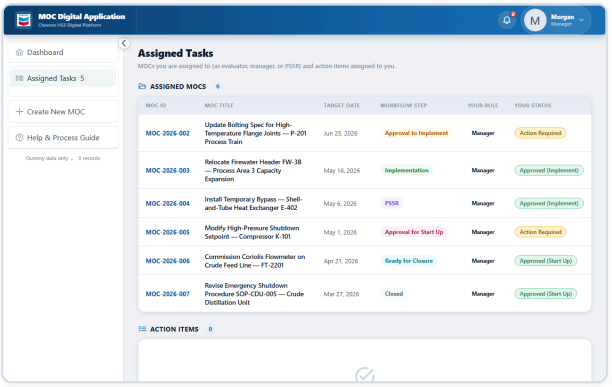
All Users

Assigned Tasks shows every MOC a user has ever been involved in — not just active ones. The **YOUR STATUS** column shows each person's personal involvement status, colour-coded for quick reading.

STATUS BADGE	WHO SEES IT	MEANING
Action Required (amber)	Manager	Awaiting your approval decision
Evaluation Complete (green)	Evaluator	Your discipline evaluation is done
Evaluation Pending (blue)	Evaluator	Your evaluation is not yet complete
Approved (Implement / Start Up) (green)	Manager	You approved this gate
Rejected (red)	Manager	You rejected — MOC is with Owner for revision
Action Items Open / Complete (blue/green)	Any user with AIs	Your action item status on this MOC



Evaluator full history — past and present MOCs with Your Status



Manager full history — approval decisions tracked across all MOCs

5. Action Item Types & Further Action Items

Action items have a **Phase** that controls when they can be raised and when they must be closed. Choosing the correct phase is critical for correct stage-gate enforcement.

PHASE	WHO CAN RAISE IT	COMPLETABLE DURING	MUST BE CLOSED BEFORE	PRIORITY LEVELS
Evaluation	Any Evaluator (during Evaluation stage)	Evaluation stage	Owner submits for Approval to Implement	Critical / High / Normal
Pre-Startup	Evaluators (Evaluation) or Operations (PSSR)	PSSR stage	Operations submits PSSR for Review	Critical / High / Normal
Post-Startup	Evaluators (Evaluation) or Operations (PSSR)	Ready for Closure stage	Owner closes the MOC	Critical / High / Normal

Further Action Items

⇗ **Further Action Items** allow a person to raise a linked action item for another person when they need additional help completing their assigned work. The further AI can be any phase type and assigned to any user.

Step 1 — Raise: Click **Raise Further AI** on any open action item assigned to you. Fill in AI type, assignee, priority, due date, description, and a comment. Click Submit.

Step 2 — Close your original: Your original action item is *not* automatically closed. A reminder badge reads "Further AI Raised — close this item yourself." You must close it normally (add a closure comment and click Complete).

Stage Gate Rule: Both the original action item *and* the further action item must be closed before the stage can advance. Neither automatically substitutes for the other.

Linking: The original item shows the further AI ID; the further AI shows "Further AI from [parent ID]". This creates a visible delegation chain in the audit trail.

6. Role & Visibility Matrix

Capability	Owner	Evaluator	Manager	Admin
Dashboard — My MOCs	Own MOCs	Own MOCs	Own MOCs	All MOCs
Assigned Tasks — MOCs	—	All assigned (full history)	All involved (full history)	All MOCs
Create MOC	✓	✓	✓	✓ (+ owner selector)
Complete evaluation checklist	—	Own discipline only	—	✓
Raise action items	—	✓	—	✓
Close action items	—	Own AIs only	—	✓
Submit for Approval to Implement	✓	—	—	✓
Approve / Reject at gates	—	—	✓	✓
Revise rejected MOC	✓	—	—	✓
Mark Implementation Complete	✓	—	—	✓
Complete PSSR checklist	—	Operations only	—	✓
Close MOC	✓	—	—	✓
Checklist Admin (configure)	—	—	—	✓
Demo Reset	—	—	—	✓

7. Key Rules & Guardrails

Assigned person only — Action items can only be completed by the person they are assigned to. No exceptions, except Admin override.

Stage-gate enforcement — Evaluation items: only completable during Evaluation. Pre-Startup items: PSSR stage only. Post-Startup items: Ready for Closure stage only. The system enforces this — buttons are hidden outside the permitted stage.

Further Action Items — manual closure required — Raising a Further AI does not auto-close the original. The person who raised the Further AI must close their original item themselves. Both must be closed before stage can advance.

Mandatory rejection reason — Manager rejection at either gate (Approval to Implement or Approval for Start Up) requires a written reason. The Submit button remains disabled until a reason is entered.

Rejection preserves full history — Every Manager rejection creates a Revision History entry on the MOC. When the Owner resubmits, all field changes are diffed (old → new) and permanently recorded. All evaluations reset on resubmission.

Owner controls advancement — After Manager approval and after PSSR completion, the Owner must manually advance the MOC. Nothing moves forward automatically at these gates.

Full audit trail — All state changes, approvals, rejections, revisions, checklist responses, and action item closures are timestamped with user identity in the Routing Tracker. The closed MOC record is immutable.

8. Technology Stack

FRONTEND FRAMEWORK
Angular 18 — single standalone component, no routing, no services

UI COMPONENT LIBRARY
PrimeNG 17 — tables, modals, tooltips, panel menus

STATE / PERSISTENCE
localStorage only — no backend, no database required

BUILD TOOL
Angular CLI — `npx ng build` → static files in `dist/`

STATIC FILE SERVER
`serve-dist.js` — Node.js server, port 4300

SCREENSHOT AUTOMATION
Playwright — `take-screenshots.js` captures 39 screenshots

Project Files

FILE	DESCRIPTION
`src/app/app.ts`	All application logic (~3 000 lines) — components, state, workflow methods
`src/app/app.html`	All views and templates (~1 800 lines)
`src/app/app.css`	All styles (~2 400 lines)
`serve-dist.js`	Static file server for the production build
`take-screenshots.js`	Playwright script — regenerates all 39 demo screenshots
`generate-pdf.js`	This script — generates official PDF documentation
`ui-screenshots/`	39 auto-generated screenshots (referenced in README and this PDF)

Regenerating Screenshots & PDF

COMMAND	PURPOSE
`npx ng build`	Rebuild app after code changes
`node serve-dist.js`	Start server on port 4300
`node take-screenshots.js`	Retake all 39 screenshots
`node generate-pdf.js`	Regenerate this PDF document